Sharp asymptotic separation of opposing egalitarian core selections

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Abstract

For singleton-normalized exact transferable-utility games on n players, let s_n be the supremum of the normalized ℓ_1 distance between the leximin and leximax-min core allocations. We prove

$$\lim_{n\to\infty} n(2-s_n) = 2, \qquad 2 - s_n = \frac{2}{n} + O\left(\frac{\log n}{n^2}\right).$$

The upper bound on separation is a short convexity argument. The matching construction is the minimum of two additive games, and its entire core is a line segment. Complementary coalition equalities encode a Fibonacci recurrence that determines the segment exactly. Both full selectors are unique at their first optimization stages. A separate three-generator construction reaches the same asymptotic constant and extends PXC151. Conventional proofs and finite exact computations are distinguished. External correctness, novelty, and significance assessment remain pending.

1 Statement and scope

Write $y(S) = \sum_{i\in S} y_i$ and

$$\mathrm{Core}(v) = \{y \in \mathbb{R}^N : y(N) = v(N),\ y(S) \geq v(S)\ (S \subseteq N)\}.$$

A game is *exact* if each coalition value is attained by a core allocation. Singleton normalization means $v(\{i\}) = 0$ for every player, so every core allocation is nonnegative. Throughout $T = v(N) > 0$. The allocation $x = \mathrm{Lmin}(v)$ lexicographically maximizes the increasing rearrangement of payoffs; $z = \mathrm{Lmax}(v)$ lexicographically minimizes their decreasing rearrangement. These are opposing egalitarian selections, not the coalition-excess nucleolus. Put

$$O(x,z) = \sum_i \min(x_i, z_i), \qquad \rho(v) = \frac{\|x-z\|_1}{T} = 2 - \frac{2O(x,z)}{T}, \qquad s_n = \sup_v \rho(v),$$

where the supremum ranges over the stated exact games.

Theorem 1. *For $n \geq 2$, every such game satisfies $\rho(v) \leq 2 - 2/(n-1)$. For every integer $n \geq 128$, writing $L = \lceil \log_2 n \rceil$,*

$$\frac{2}{n-1} \leq 2 - s_n \leq \frac{2}{n-12L} + \frac{2}{n^2+1}. \tag{1}$$

Consequently $\lim_{n\to\infty} n(2-s_n) = 2$ and $2 - s_n = 2/n + O(\log n/n^2)$. The constructions use two nonnegative efficient generators and can be scaled to integer-valued games. Fixed-n sharpness and uniqueness of asymptotic extremizers are not asserted.

Thus PXC151's coefficient $3 + 2\sqrt{2}$ is not globally optimal.

2 A generic overlap bound

Lemma 2. *Let K be a nonempty compact convex subset of $\{y \geq 0 : y(N) = T\}$, with $n \geq 2$. If x maximizes the minimum coordinate and z minimizes the maximum coordinate on K, then $O(x, z) \geq T/(n-1)$.*

Proof. If $x = z$, the claim is immediate. Otherwise put $a = \min_i x_i$ and $M = \max_i z_i$; equal totals imply $a < M$. Some index i with $x_i = a$ has $z_i \leq a$. Otherwise a sufficiently short move from x toward z raises every minimum coordinate, while the finitely many other coordinates remain above a, contradicting maximin optimality. Similarly some index j with $z_j = M$ has $x_j \geq M$. The indices are distinct. Therefore

$$O(x, z) \geq z_i + M, \qquad T \leq z_i + (n-1)M \leq (n-1)(z_i + M) \leq (n-1)O(x, z).$$

□

Neither exactness nor later lexicographic stages are needed. This reconstructs and slightly strengthens the universal bound reported in the previous session; it does not certify that session's lost release run.

3 Realizing a segment by coalition equalities

Lemma 3. *Let A be an invertible $p \times p$ zero-one matrix such that $d = A^{-1}\mathbf{1} > 0$. Choose a positive integer r with $H = r - \sum_i d_i \geq 1$. On $p+1$ donor and r receiver players, set*

$$U = (H, d_1, \ldots, d_p; 0^r), \quad V = (0^{p+1}; 1^r), \quad v(S) = \min\{U(S), V(S)\}.$$

Then v is exact and singleton-normalized, with $\mathrm{Core}(v) = [U, V]$. *If $\alpha = \min_i d_i$, its full selectors are $tU + (1-t)V$ at*

$$t_{\min} = \frac{1}{1+\alpha}, \qquad t_{\max} = \frac{1}{1+H}, \qquad 2 - \rho(v) = 2\left(\frac{\alpha}{1+\alpha} + \frac{1}{1+H}\right).$$

Proof. Both nonnegative generators have total r, belong to the lower-envelope core, and collectively attain every coalition value. Their disjoint supports give singleton normalization. Each column of A occurs in a row, so $Ad = \mathbf{1}$ and positivity imply $d_i \leq 1$.

Let D be the donor subset of any row of A. For each receiver e, both generators give $D \cup \{e\}$ value one and its complement value $r-1$. The two core inequalities and efficiency force $y(D) + y_e = 1$. All receivers thus have a common payment $1 - t$. The other donor coordinates satisfy $Ay_d = t\mathbf{1}$, hence $y_d = td$; efficiency gives $y_h = tH$. Nonnegativity implies $0 \leq t \leq 1$. Convexity gives the reverse containment.

On this segment, the minimum coordinate is $\min(t\alpha, 1-t)$ and the maximum is $\max(tH, 1-t)$, since $0 < \alpha \leq d_i \leq 1 \leq H$. Each scalar optimum is unique, so already determines the full selector. The endpoint supports are disjoint with equal total, giving normalized distance $2(t_{\min} - t_{\max})$ and the formula. □

The structural content is preserving a segment under coalition closure; an arbitrary convex segment need not be the core of its lower envelope.

4 The Fibonacci family

Let $F_1 = F_2 = 1$, $F_{j+2} = F_{j+1} + F_j$. For integers $m \geq 2$, $r \geq m+1$, write

$$F = F_{m+1}, \quad H = r - m, \quad a_i = F_{m-i+1}/F, \quad b_i = 1 - a_i \quad (1 \leq i \leq m).$$

Define

$$U = (H, a_1, \ldots, a_m, b_1, \ldots, b_m; 0^r), \qquad V = (0^{2m+1}; 1^r). \tag{2}$$

There are $n = r + 2m + 1$ players and total $T = r$.

Proposition 4. *The lower envelope of* (2) *has core exactly* $[U, V]$*, and*

$$\rho = 2\left(\frac{F}{F+1} - \frac{1}{r-m+1}\right), \qquad 2 - \rho = \frac{2}{F_{m+1}+1} + \frac{2}{r-m+1}. \tag{3}$$

Proof. With donor labels A_i, B_i, these $2m$ subsets have weight one:

$$\begin{aligned} &\{A_i, B_i\} && (1 \leq i \leq m),\\ &\{A_{i+1}, A_{i+2}, B_i\} && (1 \leq i \leq m-2),\\ &\{A_{m-1}, B_m\}, \qquad \{A_1, A_2\}. \end{aligned}$$

Their incidence matrix is invertible. The homogeneous equations eliminate the B_i variables and give $A_i = A_{i+1} + A_{i+2}$, $A_{m-1} = A_m$, and $A_1 + A_2 = 0$. Writing $A_m = u$ gives $A_i = F_{m-i+1}u$ and $F_{m+1}u = 0$. Thus all variables vanish. This includes $m = 2$, when there are no recurrence rows. The positive inhomogeneous solution (a, b) has total m and minimum $1/F$. Apply Lemma 3. □

Proof of Theorem 1. Lemma 2 gives the lower bound on $2 - s_n$. For $n \geq 128$ choose $L = \lceil \log_2 n \rceil$, $m = 4L$, and $r = n - 2m - 1$. These satisfy $r \geq m + 1$: at $n = 128$ check directly; for $L \geq 8$, $n \geq 2^{L-1} + 1 \geq 12L + 2$. The recurrence implies $F_{2j+1} \geq 2^j$ for $j \geq 0$. Hence $F_{m+1} \geq 2^{2L} \geq n^2$ and $r - m + 1 = n - 12L > 0$. Formula (3) proves (1). Since $L = O(\log n)$, subtracting $2/n$ and squeezing proves the claims. □

Multiplication by F makes all generator coordinates integral. Choosing instead the least $m \geq 2$ with $F_{m+1} \geq n^2$ gives $m = O(\log n)$ and $n^2 \leq F_{m+1} < 2n^2$ for large n. The scaled total is $rF = O(n^3)$, so every coordinate needs only $O(\log n)$ bits.

5 An independent extension of PXC151

A separate three-generator construction and full proof are included in `THREE_GENERATOR_PROOF.md`. For integers $k \geq 1$, $m \geq \max(2, 2k-2)$, $r \geq 3$, put $q = (k+1)m - k + 2$. Its $n = m + k + r + 3$ players have total $T = rq + m + 1$, both optimal first-stage faces are singletons, and its exact overlap is $O = q + r + 1$. Thus

$$\rho = 2 - \frac{2(q + r + 1)}{rq + m + 1}.$$

For fixed k, optimizing m/r gives coefficient $2(1 + 1/\sqrt{k+1})^2$. At $k = 1$ this is PXC151; it decreases to two as k grows. The polynomial sequence $k = s^2, m = 2s^2, r = 2s^3$ gives $n(2 - \rho) \to 2$. This family increases the multiplicity of an existing payoff role, and uses a different mechanism from the Fibonacci segment. Its domain restriction is material: an exact negative control at $(m, k, r) = (3, 3, 3)$ falsifies the extrapolated leximin formula.

6 Verification and review boundary

The main theorem has a conventional all-parameter proof. A standalone rational Z3 replay reconstructs all coalition inequalities for five Fibonacci cases with $8 \le n \le 14$, proves that the whole core is the segment, and separately verifies both optimum values and uniqueness. A perturbed unit-sum system produces an off-segment core allocation, checking that coalition closure can enlarge a perturbed segment. Fibonacci identities are also calibrated for $2 \le m \le 64$; this is finite evidence, not an all-parameter proof. The separate three-generator replay checks all coalitions for four cases with $n = 9, 10, 12, 13$, excluding better floors, better ceilings, and distinct allocations on either optimal face. A parameter-domain falsifier gives an explicit improved core point. The generic near-extremizer's coalition closure is independently calibrated in 24 cases.

These fairness selections are established concepts studied algorithmically in assignment-game cores and Owen sets [1, 2]. Classical exactness and lower envelopes are not asserted as novel. Exact-cone facets and majorization characterizations provide adjacent structural frameworks [3, 4]. The candidate contribution is the sharp asymptotic separation theorem and its explicit realizations. A targeted primary-source screen did not locate an equivalent quantitative result; that is not a novelty certification. Specialist correctness, prior-art, and significance review remain pending.

References

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